

Department of Mechanical Engineering
University of South Carolina
Columbia, SC 29208
January 30, 2026

Dear Members of the Evaluation Committee,

I am recommending Sam Cancilla for the SC Space Grant Undergraduate Research Award. I have supervised Sam in the Integrated Multiphysics and Systems Engineering Laboratory (iMSEL) since March 2025, and he is among the most capable and dedicated undergraduates I have worked with in my eight years at the University of South Carolina.

What sets Sam apart is that he owns problems end-to-end. I gave him the open-ended task of developing a radar-based tracker for autonomous maritime perception. Rather than waiting for a scripted plan, he scoped the data requirements with a graduate student, organized radar collection campaigns across three South Carolina lakes and Charleston Harbor – over 38 hours total – and built the full detection-and-tracking pipeline from scratch. During one campaign at Lake Greenwood, unexpected wind gusts changed the clutter environment mid-collection; he revised the collection settings on-site to capture both regimes, turning a setback into a richer dataset.

His technical judgment is equally strong. He implemented the tracking pipeline in Rust for an embedded NVIDIA Jetson Orin platform, and when profiling revealed memory-allocation bottlenecks, he independently redesigned the pipeline around pre-allocated buffers before raising it at our group meeting – arriving with a working solution, not just a problem. He is now co-author on a paper submitted to AIAA, and the algorithm presented in that paper is entirely his work. Most undergraduates contribute to existing codebases; Sam created one.

His proposed research – extending adaptive clustering to temporal self-tuning and adding a vessel classifier – is well scoped. He completed the current pipeline from design through deployment and ablation testing in 10 weeks; the proposed extensions are narrower, and the 14-week timeline is realistic.

Beyond the lab, Sam serves as an Army National Guard 09R SMP Cadet through ROTC and as Assistant Opinion Editor of The Daily Gamecock, maintaining a 3.5 GPA while balancing military obligations, editorial deadlines, and fieldwork. The operational discipline he brings – equipment checklists, field logs, post-campaign debriefs – is something the team relies on when planning data collection.

I give Sam my strongest recommendation for this award.

Sincerely,

Dr. Yi Wang
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